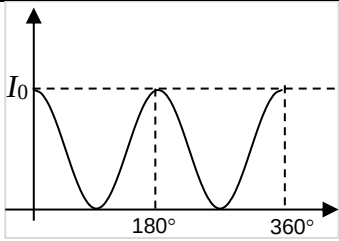
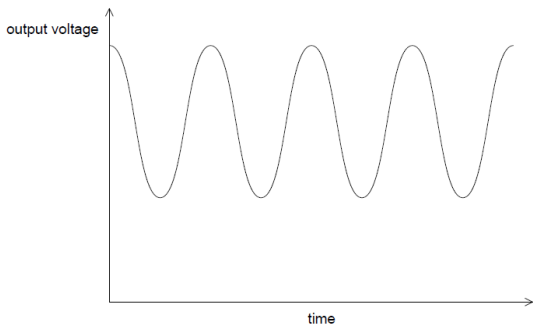


6(a)(i)	The light is polarised in the same direction as the filter's polarising direction / polarised in the vertical direction.
(a)(ii)	
(a)(iii)	$\cos \theta = \sqrt{0.179} \rightarrow \theta = 65.0^\circ$ $I_0 \cos^2 (65.0^\circ - 30^\circ) = 0.67 I_0$
(b)(i)	Light from both slits arrive in phase, it would result in constructive interference; if arrive in π rad out of phase, it would result in destructive interference. This corresponds to maximum voltage for constructive and minimum for destructive. As the train moves, it would detect alternating high and low voltages / interference pattern. Peak decreases due to diffraction.
(b)(ii)1.	$x = \frac{\lambda D}{d}$ $= \frac{(630 \times 10^{-9})(5.0)}{1.5 \times 10^{-3}}$
	$x = 2.1 \times 10^{-3} \text{ m}$
(b)(ii)2.	Time between two peaks = $\frac{0.01}{5} = 2 \times 10^{-3} \text{ s}$
	$v = \frac{2.1 \times 10^{-3}}{2 \times 10^{-3}}$ $v = 1.05 \text{ m s}^{-1}$
(c)(i)	$\frac{\lambda}{b} = \frac{0.13}{5.0}$ $b = (630 \times 10^{-9}) \left(\frac{5.0}{0.13} \right)$
	$b = 2.4 \times 10^{-5} \text{ m}$
(c)(ii)	Peak voltages will be higher. Width of the central bright fringe will be smaller.
(d)	
	Reflected waves and incidence waves interfere.

	Same frequency, opposite direction, hence a stationary wave is formed.